

E-commerce Profit Leakage & Recovery Analysis Dashboard

Complete Project Explanation | Advanced Excel Portfolio Project

Project Type	Advanced Excel Data Analytics Project
Business Theme	E-commerce profit leakage, return risk, COD risk, and recovery analysis
Main Tool	Microsoft Excel
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1. Project Story in Simple Words

Imagine an e-commerce company is getting good sales and many orders. At first, the business looks healthy because revenue is high. But when the company checks actual profit, it realizes that profit is not growing as expected.

This usually happens because money is leaking in hidden areas such as high discounts, product returns, COD risk, shipping cost, and low-margin products.

This project was built to answer one main question:

Where is the business losing money, and how much profit can be recovered?

The project is not just a sales dashboard. It is a profit leakage, return risk, COD risk, product leakage, and profit recovery analysis system built in Excel.

2. Business Problem and Objective

Many beginner dashboards only show revenue and orders. But in real business, revenue alone is not enough. A company can have high revenue and still have weak profit if the business is losing money through discounts, returns, or high operating costs.

This project focuses on finding those hidden leakage points and converting them into business recommendations.

Main business questions answered by this project:

- Where is profit leaking?
- Which categories, cities, and products are causing losses?
- Which payment methods are risky?
- How much money is lost due to discounts and returns?
- How much profit can be recovered if business actions are taken?
- What should management do next?

3. Workbook Structure Overview

The workbook follows a professional analysis flow. Raw order data is stored in one place, backend calculations are separated from the dashboard, and the final dashboard presents insights to business users.

Step	Sheet / Area	Role
1	Raw_Data	Stores 1,000 order-level records. This is the single source of truth.
2	Calculations	Calculates KPIs and analysis tables. This is the backend engine.
3	Analysis Sheets	Return risk, product leakage, pivot analysis, and chart data.
4	Dashboard	Shows final KPIs, insights, charts, and recommendations.
5	Profit_Recovery_Sim	Estimates possible monthly and yearly profit recovery.

4. Sheet-by-Sheet Explanation

README

This sheet works like project documentation. It explains the project overview, business problem, tools used, sheet guide, key insights, and final recommendations. It helps any new viewer understand the workbook without asking the creator.

Raw_Data

This is the foundation of the project. It contains 1,000 e-commerce orders. Each row represents one order. All calculations and dashboards are connected directly or indirectly to this sheet.

Calculations

This is the engine room of the workbook. It calculates KPIs such as revenue, orders, gross profit, net profit, discount leakage, return rate, COD ratio, shipping cost, return loss, and average order value.

Profit_Recovery_Sim

This sheet works like a what-if simulator. It estimates how much profit can be recovered if the company reduces discount, reduces returns, optimizes shipping cost, removes low-margin products, or converts COD users to prepaid/UPI.

Return_Risk_Analysis

This sheet identifies high-risk orders. It uses risk score logic based on discount and COD behavior. It helps the operations team identify orders that may need extra verification before dispatch.

Product_Analysis

This sheet identifies top loss-making products and products with high discount leakage. It helps the business decide which products should be repriced, monitored, bundled, or discontinued.

Charts_Data

This sheet stores clean chart-ready summary tables. It makes dashboard charts easier to manage and keeps visualizations separate from raw data.

Pivot_Analysis

This sheet uses a Pivot Table to validate payment method risk. It shows that the project uses both formula-based analysis and pivot-based analysis.

Dashboard

This is the final executive dashboard. It presents KPI cards, analysis tables, charts, key insights, and business recommendations in one place.

5. Raw_Data Column Explanation

Raw_Data is the main dataset. Each column has a business meaning. Some columns are raw input columns, and some are calculated columns used for analysis.

Column	Meaning	Why it matters
Order_ID	Unique order number	Helps count and track orders.
Date / Month	Order date and month	Used for monthly trend analysis.
Category / Sub_Category	Product grouping	Used for category-level performance analysis.
Product_Name	Name of product sold	Used for product-level leakage analysis.
City / Region	Customer location	Used for city and region risk analysis.
Units_Sold	Quantity sold	Used to calculate revenue and profit.
MRP	Original price	Used to calculate discount leakage.
Selling_Price	Actual selling price	Used to calculate revenue.
Discount_Pct	Discount percentage	Helps analyze discount risk.
Cost_Price	Company cost	Used to calculate gross profit.
Gross_Profit	Profit before shipping	Shows product profitability before delivery cost.
Shipping_Cost	Delivery cost	Reduces final profit.
Net_Profit	Profit after shipping	Shows actual final profit.
Payment_Method	COD, UPI, prepaid, etc.	Used for payment risk analysis.
Order_Status	Delivered, returned, cancelled, etc.	Used to measure returns and cancellations.
Return_Flag	1 for returned, 0 otherwise	Makes return counting easy.
COD_Flag	1 for COD, 0 otherwise	Makes COD risk analysis easy.
Return_Risk_Score	Calculated risk score	Identifies high-risk orders.

6. Important Formulas and Logic

This section explains the most important formulas in simple English. These formulas convert raw order data into business KPIs and insights.

Gross Profit

`= (Selling_Price - Cost_Price) * Units_Sold`

This calculates profit before shipping cost. First, selling price minus cost price gives profit per unit. Then it is multiplied by units sold.

Net Profit

`= Gross_Profit - (Shipping_Cost * Units_Sold)`

This calculates final profit after delivery cost. Shipping cost reduces actual profit.

Return Flag

```
=IF(Order_Status="Returned",1,0)
```

This marks returned orders as 1 and all other orders as 0. It makes return counting simple.

COD Flag

```
=IF(Payment_Method="COD",1,0)
```

This marks COD orders as 1 and other payment methods as 0. It helps analyze COD risk.

Return Risk Score

```
=MIN(1,(Discount_Pct*1.4)+(COD_Flag*0.35))
```

This score increases when discount is high and when the order is COD. MIN keeps the score from going above 1.

Total Revenue

```
=SUMPRODUCT(Selling_Price, Units_Sold)
```

SUMPRODUCT multiplies selling price by units sold for each order and then adds the total.

Discount Leakage

```
=SUMPRODUCT(MRP - Selling_Price, Units_Sold)
```

This calculates total money lost because products were sold below MRP.

Return Rate

```
=Returned Orders / Total Orders
```

This shows what percentage of orders were returned.

COD Order %

```
=COD Orders / Total Orders
```

This shows how much business is coming from COD orders.

Category Net Profit

```
=SUMIFS(Net_Profit, Category, Selected_Category)
```

This adds net profit for a specific category only.

City Return Rate

```
=City Returns / City Orders
```

This shows which cities have higher return problems.

Top Risky Orders

```
=TAKE(SORTBY(CHOOSECOLS(...), Return_Risk_Score, -1), 30)
```

This selects important columns, sorts orders by risk score, and returns the top 30 risky orders.

7. KPI Explanation and Business Meaning

KPIs are the main business numbers shown on the dashboard. They help managers understand business performance quickly.

KPI	Meaning	Business interpretation
Total Revenue	Total sales after discount	Higher revenue is good, but it must be checked with profit.
Total Orders	Number of orders	Shows business volume.
Total Units Sold	Total quantity sold	Shows product movement.
Gross Profit	Profit before shipping cost	Shows product-level profitability.
Net Profit	Profit after shipping cost	Most important final profit measure.
Average Discount %	Average discount offered	Too high discount can reduce profit.
Return Rate %	Returned orders divided by total orders	Lower return rate is better.
COD Order %	COD orders divided by total orders	High COD share may increase risk.
Average Shipping Cost	Average delivery cost	Lower shipping cost improves margin.
Average Net Margin %	Profit percentage after costs	Higher margin is better.
Discount Leakage	Money lost due to discounts	Lower leakage means better pricing control.
Return Loss	Profit loss due to returned orders	Lower return loss improves profitability.
Revenue at MRP	Possible revenue without discounts	Useful as a comparison metric.
Average Order Value	Revenue divided by orders	Shows average customer order value.

8. Profit Recovery Simulator Explanation

The Profit Recovery Simulator is one of the strongest parts of the project. It does not only show the problem. It also estimates how much profit can be recovered if the company takes action.

Input levers used in the simulator:

- Discount Reduction % - how much unnecessary discount can be reduced.
- Target Return Rate - the improved return rate the company wants to achieve.
- Shipping Cost Reduction % - how much delivery cost can be reduced.
- Low-Margin Product Removal % - how much low-margin business can be controlled.
- COD to Prepaid Conversion % - how many COD customers can be shifted to prepaid/UPI.

The simulator calculates monthly recovery first. Then monthly recovery is multiplied by 12 to estimate annual recovery. This makes the result easier for business planning.

Example: If the business reduces discount leakage by even a small percentage every month, the annual recovery can become meaningful.

9. Return Risk Analysis Explanation

Return_Risk_Analysis works like an early warning system. It identifies orders that may become risky because of high discount, COD payment, or weak profitability.

A very good improvement in the project is showing COD High Risk as a count out of total COD orders. For example, 133 / 202 means 133 high-risk COD orders out of 202 total COD orders. This is better than showing only 133 because it gives context.

The Top 30 risky orders table helps an operations team check risky orders before dispatch. In real business, this can reduce returns and delivery losses.

10. Product Analysis Explanation

Product-level analysis is important because category-level analysis may hide exact problem products. A category may look profitable overall, but some products inside that category may be loss-making.

This sheet identifies top loss-making products and top high discount leakage products. These products should be reviewed for pricing, discount cap, bundling, stock control, or discontinuation.

11. Dashboard Walkthrough

The dashboard is the final executive view. It should be presented first to an interviewer or business manager.

Dashboard Section	What it shows	Why it matters
Title	Project name and theme	Gives clear context.
KPI Cards	Revenue, orders, profit, leakage, return, COD	Shows business health quickly.
Category Performance	Category-wise revenue, profit, returns, discount	Finds strong and weak categories.
City Analysis	City-wise profit and returns	Finds location-level issues.
Monthly Trend	Monthly revenue, profit, returns, discount	Shows business movement over time.
Payment Method Impact	COD, UPI, prepaid, cards performance	Validates payment risk.
Charts	Visual comparison of main analysis areas	Makes insights easier to understand.
Key Insights Box	Summary of findings	Saves time for the viewer.
Recommendations	Action cards	Turns analysis into business decisions.

12. Business Insights and Recommendations

Insight	Recommendation	Expected business impact
Discount leakage is high	Control unnecessary discounts and cap discounts on low-margin products	Improves net profit and pricing discipline.
Returns reduce profit	Improve product quality checks, descriptions, and return policy	Reduces return loss and operational cost.
COD orders need monitoring	Encourage prepaid and UPI payments	Reduces return and non-delivery risk.
Some products are loss-making	Review, reprice, bundle, or discontinue weak products	Improves product portfolio profitability.
Shipping cost affects margin	Negotiate logistics cost and increase free-shipping threshold	Improves final net margin.
Some cities may have higher risk	Monitor high-return cities and apply stricter controls	Reduces location-level leakage.